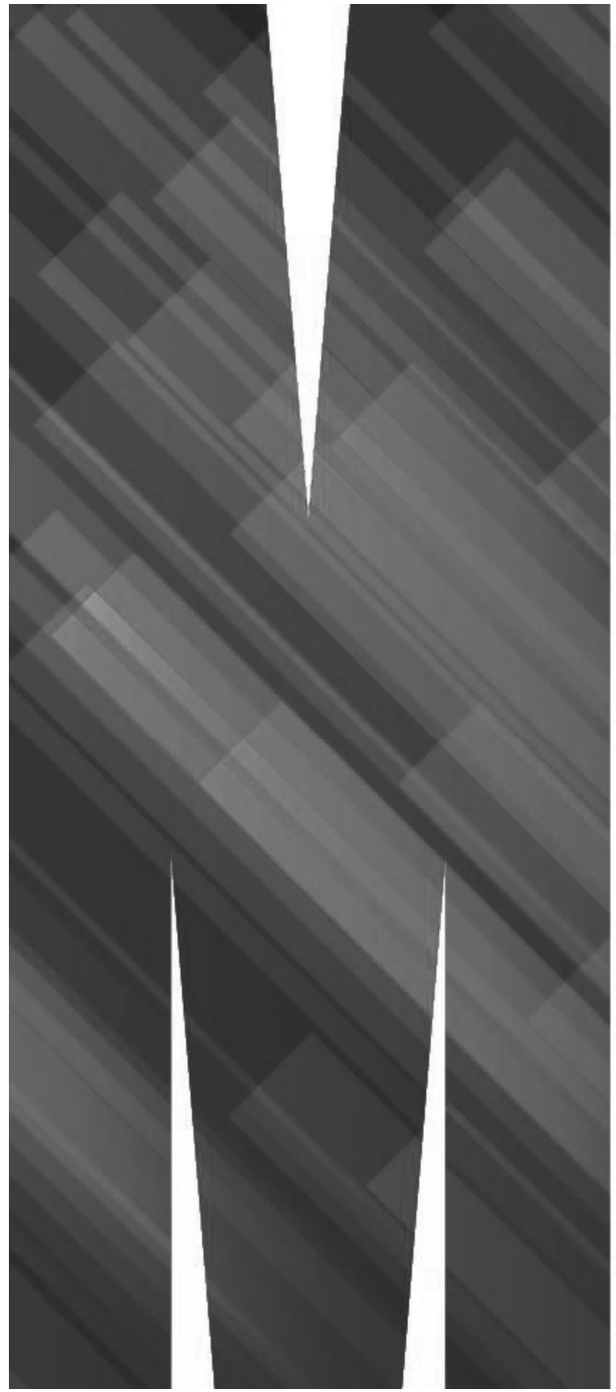
MONASH University

Topic 6.1
Regression Models

Introduction to Linear regression

By Dr Lee How Chinh



Objectives

- Describe the features of the linear regression model with its assumptions.
- Explain and apply the stepwise selection methods for selecting the linear regression inputs
- Understanding linear regression models as predictive models.

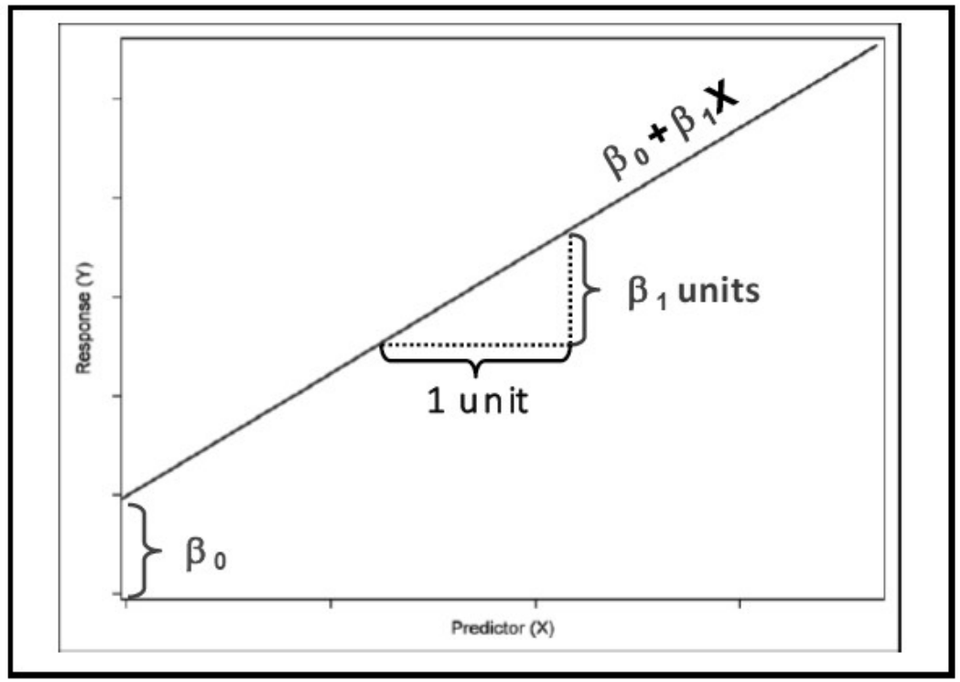
Review of Simple Linear Regression

Linear Regression Models

- Linear regression models, as parametric models, assume a specific association structure between inputs and target.
- Observations are scored or predicted using a simple mathematical prediction formula.
- The heuristic sequential selection technique is used to choose the useful inputs and generates a series of models with increasing complexity.
- Fit statistics calculated from validation data decide the champion model from the sequence.

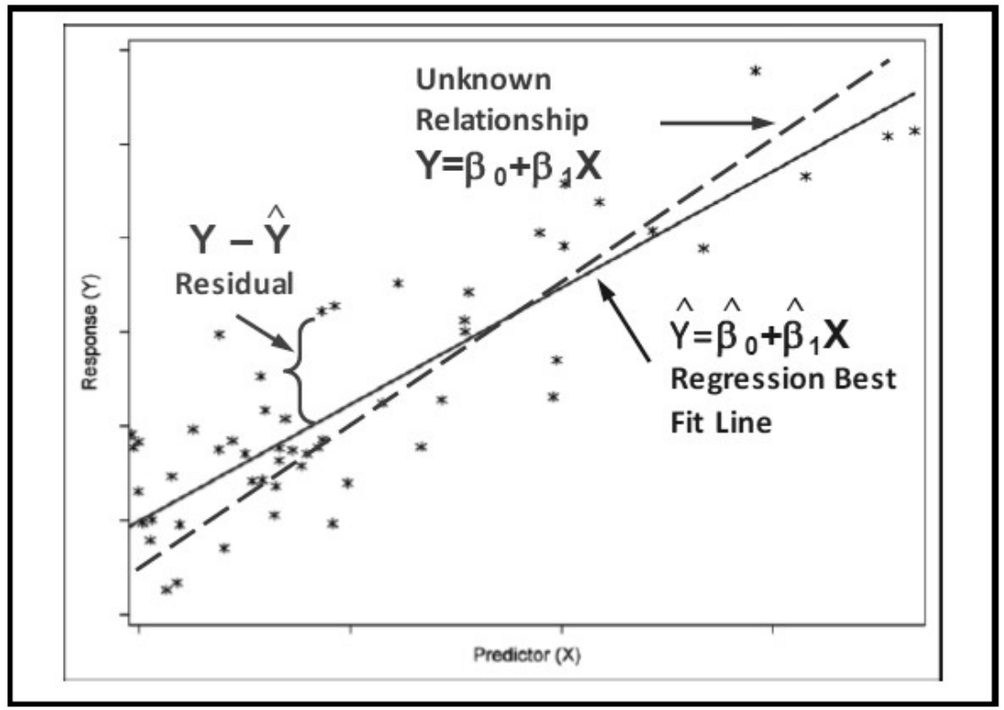
The Simple Linear Regression Model

- The relationship between the target or response variable and the input or predictor variable can be categorized by the equation $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $i = 1, \dots, n$.
- y_i is the target or response variable,
- x_i is the input or predictor variable,
- β_0 is the intercept parameter,
- β_1 is the slope parameter,
- ε_i is the error term.



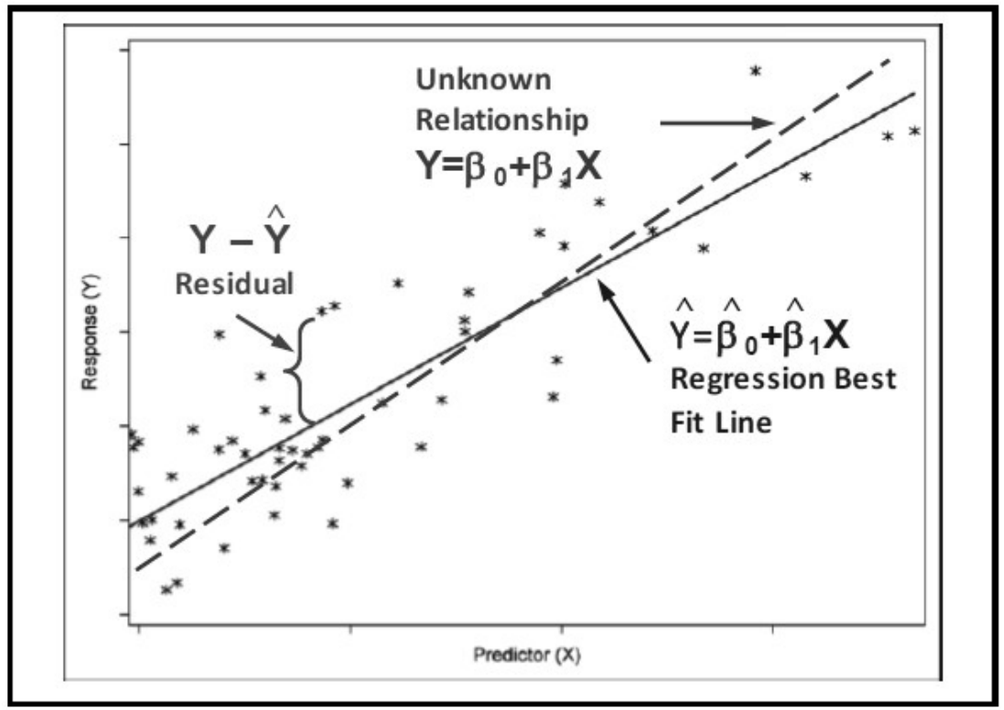
Ordinary Least Squares (OLS) Regression

- Estimates of the unknown population parameters β_0 and β_1 are obtained by the method of **ordinary least squares**.
- This method provides the estimates by determining the line that minimises the sum of the squared vertical distances between the observations and the fitted line.



Ordinary Least Squares (OLS) Regression

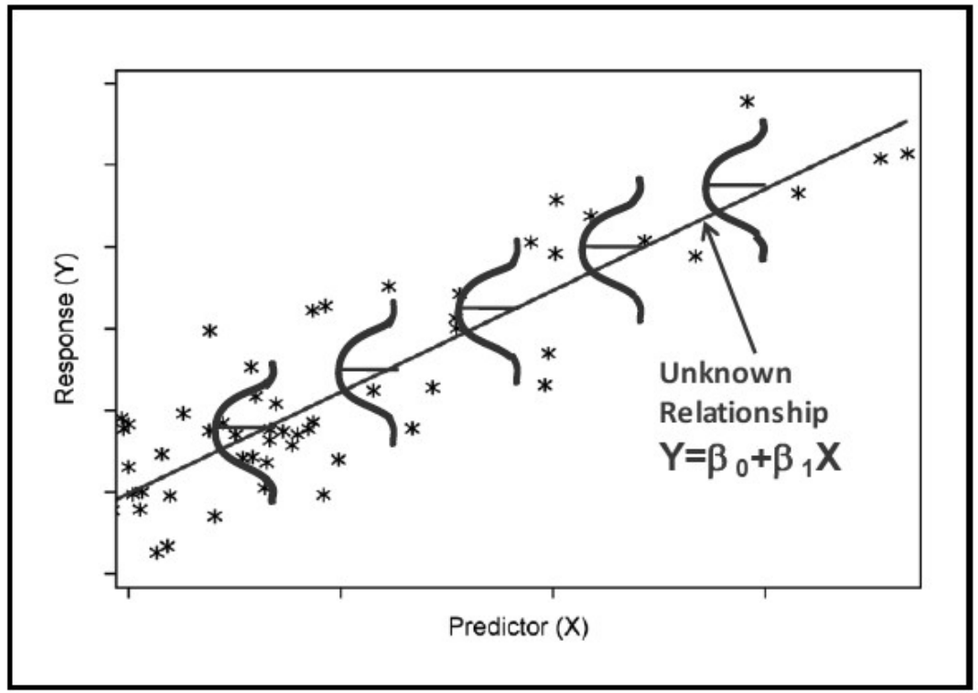
- If the assumptions of simple linear regression are valid, the least squares estimates are unbiased estimates of the population parameters and have minimum variance (efficiency).



Assumptions of simple linear regression

The following are the assumptions of simple linear regression.

- The mean of the target variable is linearly related to the value of the input variable.
- The error is normally distributed.
- The error has constant variance across the range of the input variable.
- The observations are independent.



General Linear Models

The General Linear Model

- In the general linear model (GLM), a target or response **Y** is modeled as a linear function of input or predictor variables **X**:

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where

- $\boldsymbol{Y}$ is a vector of continuous response values
- $\boldsymbol{X}$ is a matrix of predictor variables
- $\boldsymbol{\beta}$ is a vector of model parameters
- $\boldsymbol{\varepsilon}$ is a vector random errors.

The General Linear Model

- General linear models are often classified according to the types of input (predictor) variables.

Type of Predictor Variables in X		
Categorical	**Continuous**	**Continuous and Categorical**
Analysis of Variance (ANOVA)	Ordinary Least Squares (OLS) Regression	Analysis of Covariance (ANCOVA)

Assumptions of General Linear Models

- The mean of the Ys is accurately modelled by a function of the Xs that is linear in the parameters.
- The random error term, $\boldsymbol{\varepsilon}$, is assumed to have a normal distribution with a mean of zero and a constant variance, $\boldsymbol{\sigma^2}$.
- The error are independent.

$$\varepsilon \sim iid\ N(0, \sigma^2)$$

Violation of Model Assumptions

- Normality does not affect the parameter estimates, but it affects the test results.
- Constant variance does not affect the parameter estimates, but the standard errors are compromised.
- Independent observations do not affect the parameter estimates, but the standard errors are compromised.
- Linear in the parameters indicates a misspecified model, and therefore, the model results are not meaningful.

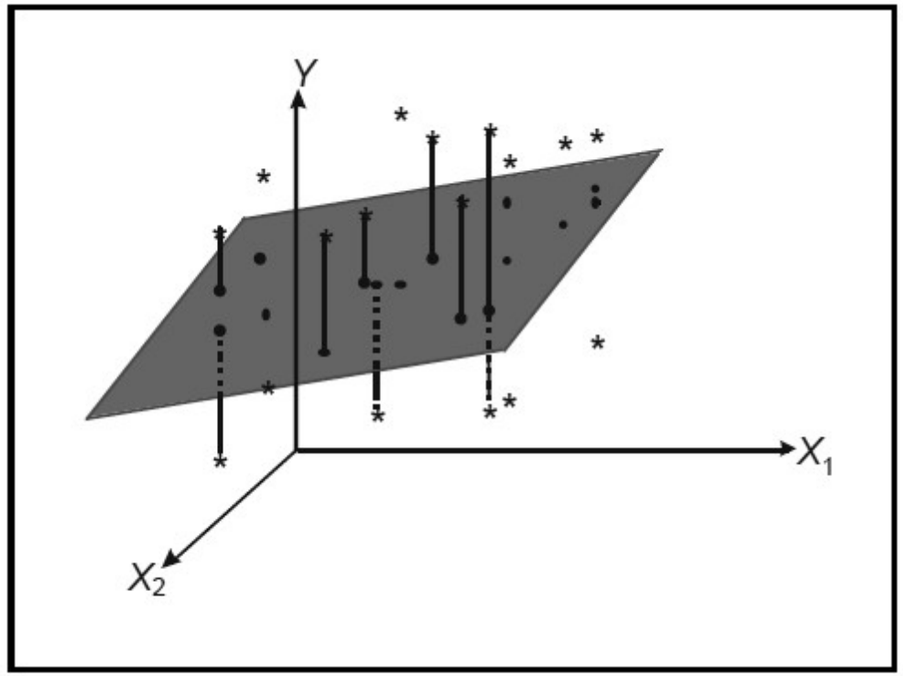
The Multiple Linear Regression Model

- In general, the multiple linear regression model can be given by the following equation

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + \varepsilon_i$$

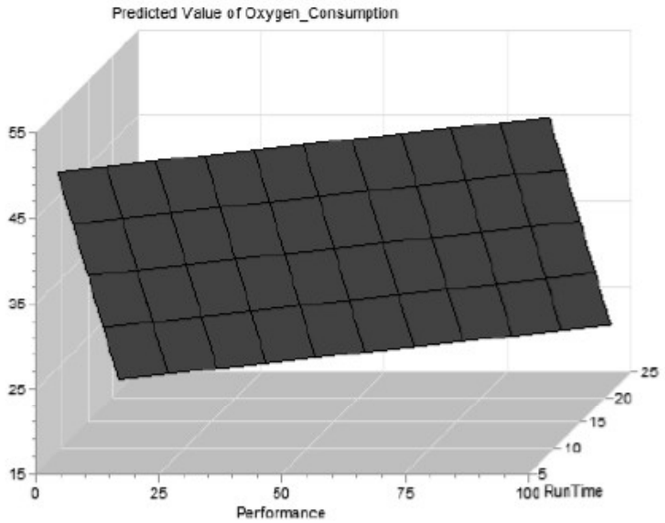
- The linear model can be written in matrix notation

$$\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon}$$



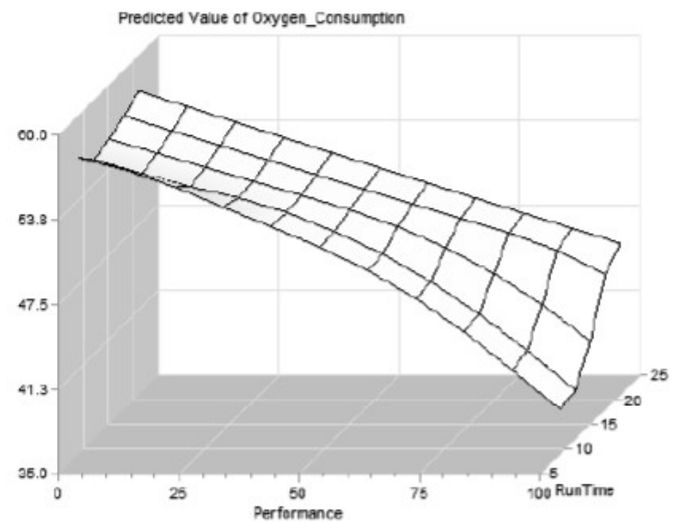
The Multiple Linear Regression Model

In general, the response variable , Y, as a **linear function** of *k* independent variables, X_1 through X_k.



$Y=\beta_0+\beta_1X_1+\beta_2X_2+\varepsilon$

Linear Model with only Linear Effects



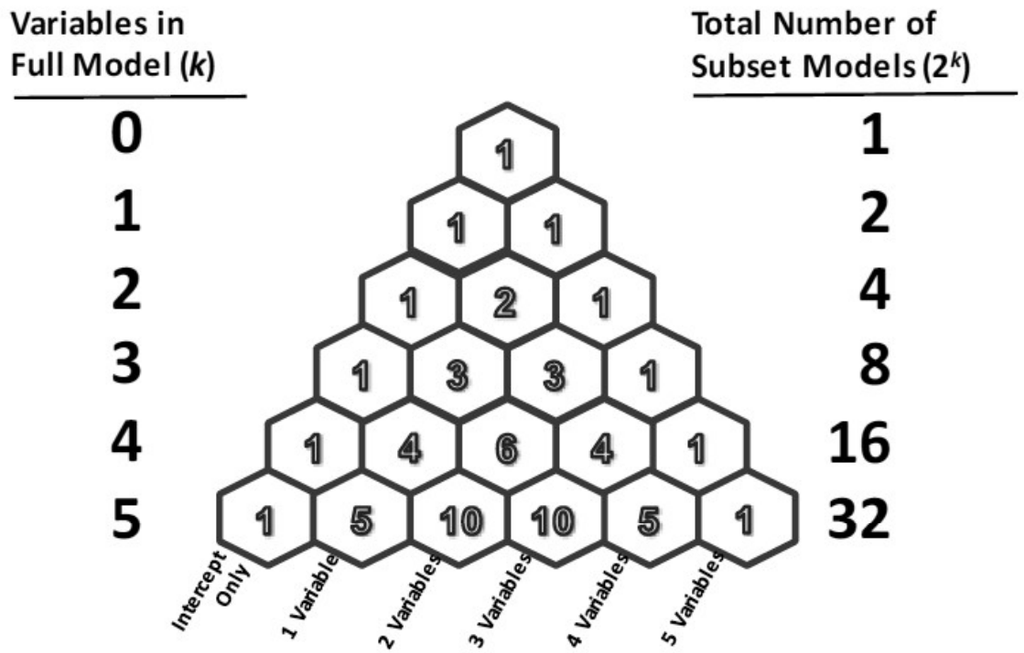
$Y=\beta_0+\beta_1X_1+\beta_2X_1^2+\beta_3X_2+\beta_4X_2^2+\varepsilon$

Linear Model with Nonlinear Effects

Sequential Selection Methods

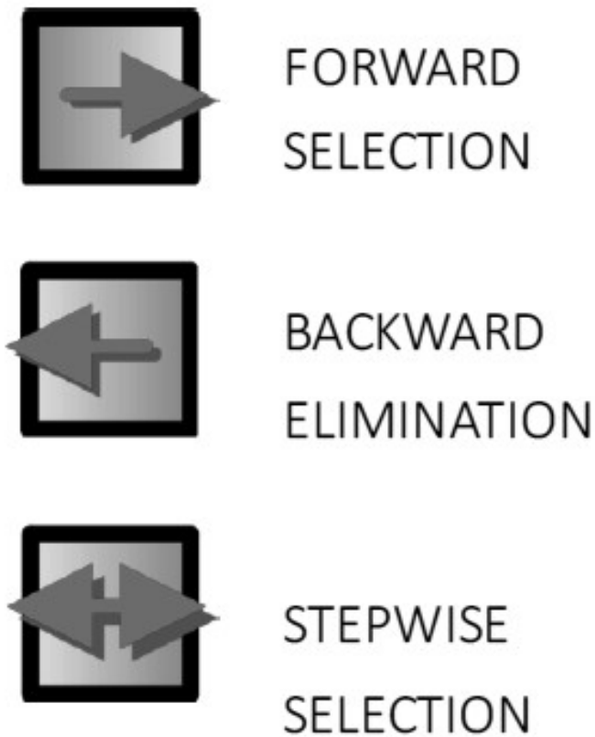
Inputs Selection Methods

- The best way to find the optimal set of inputs for a regression model is to try the combination of every input variable.
- Unfortunately, the number of models to consider using this approach increases exponentially in the number of available inputs.



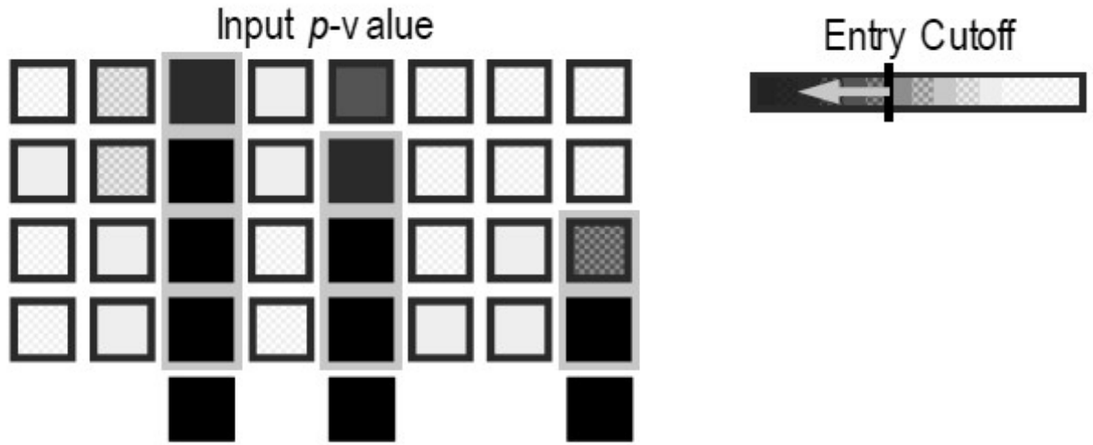
Sequential Selection Methods

- An alternative to the exhaustive search is to restrict the search to a sequence of improving models
- This method might not find the single best model, it is commonly used to find models with good predictive performance.



Forward Selection Method

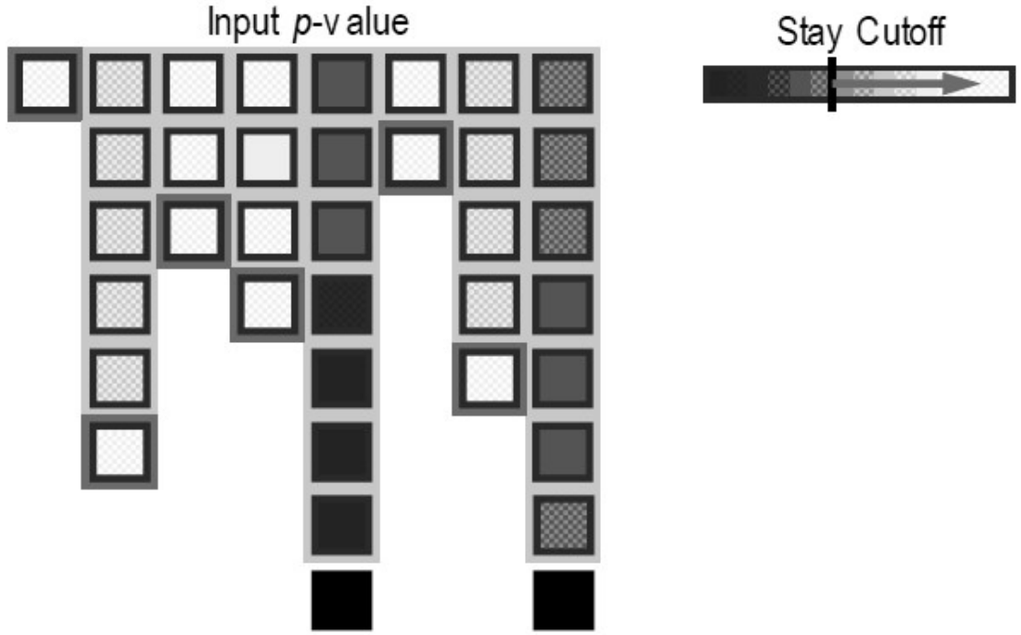
- Forward selection creates a sequence of models of increasing complexity.
- The sequence starts with the baseline model.
- First, the algorithm selects the one-input model that most improves on the baseline model.
- Then it selects the best two inputs model that contains the first selected input variable.



- The process continues until it reaches the point where no additional variables have p-value levels less than the stopping criterion.

Backward Selection Method

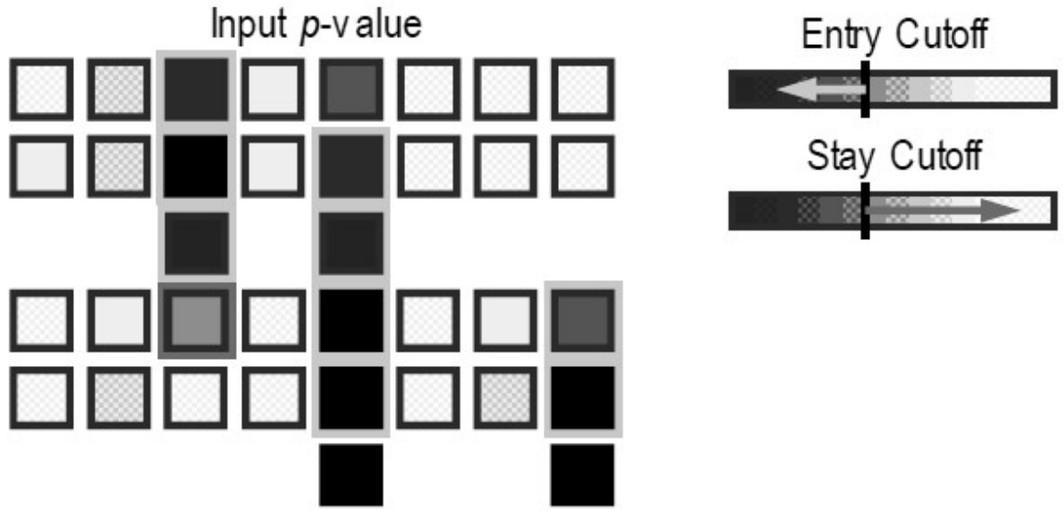
- Backward selection creates a sequence of models of decreasing complexity.
- The sequence starts with a saturated model that contains all available inputs.
- Inputs are sequentially removed from the model.
- At each step, the input selected for removal least reduces the overall model fit statistics, and this is equivalent to removing the input with the highest *p*-value.



- The process terminates when all remaining inputs have a *p*-value that is less than the stay cutoff.

Stepwise Selection Method

- Stepwise selection combines elements from both the forward and backward selection procedures.
- The method starts in the same way as the forward selection.
- After each input is added, the algorithm reevaluates the statistical significance of all included inputs. If the *p*-value of any inputs exceeds the stay cutoff, the input is removed from the model.



- The process terminates when all inputs available for inclusion in the model have *p*-values more than the entry cutoff and all inputs in the model have *p*-values less than the stay cutoff.

Shortcomings of sequential selection

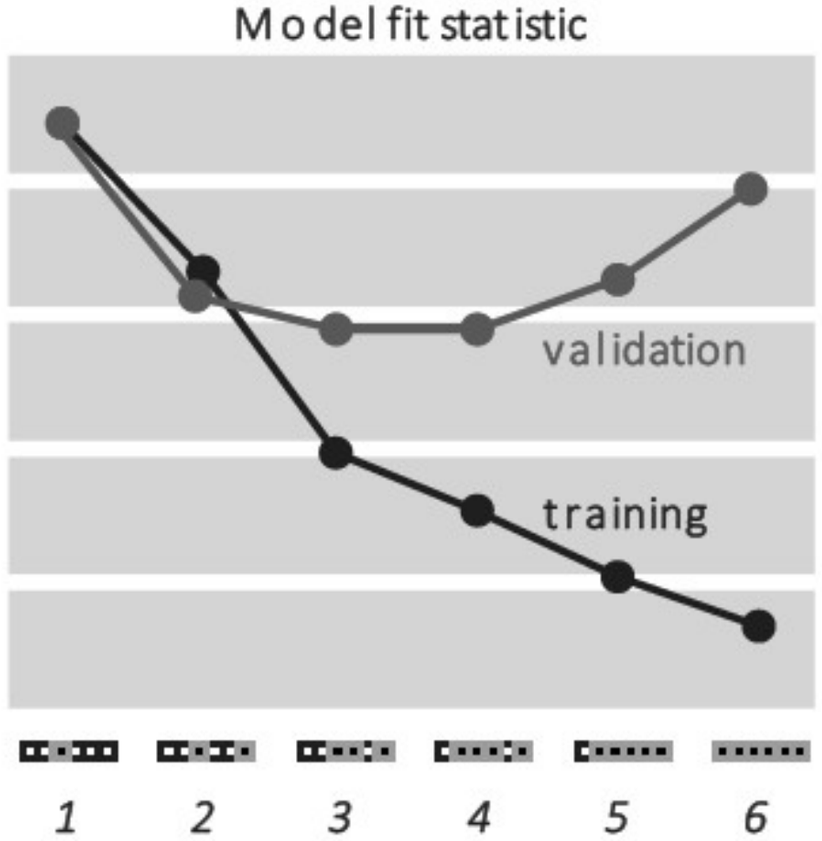
Sequential model selection results in the following:

- biases in parameter estimates, predictions, and standard errors
- incorrect calculation of degrees of freedom
- *p*-values that tend to err on the side of overestimating significance (increasing the probability of Type I error)

Best Model From Sequence

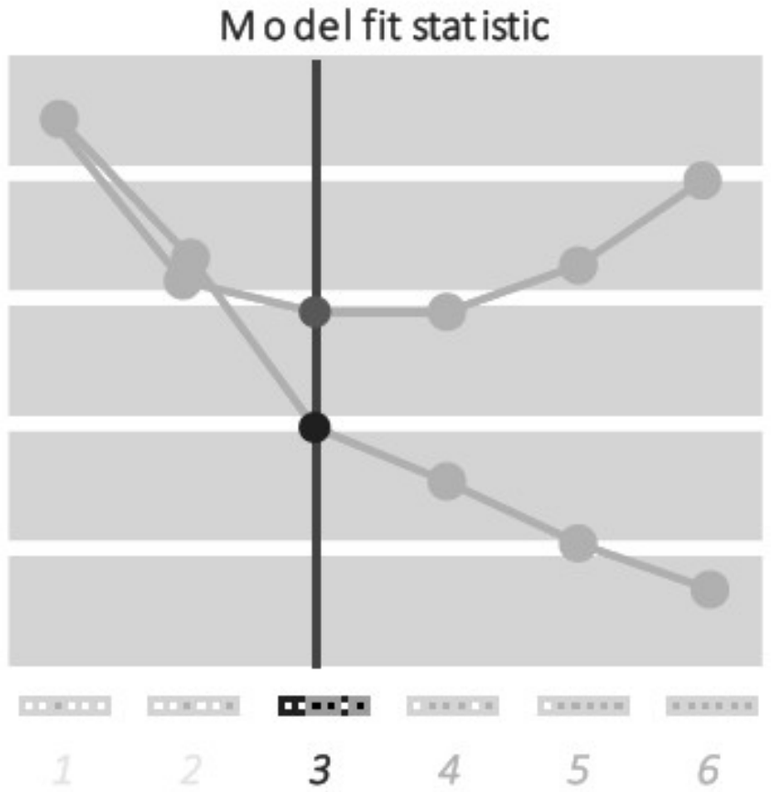
Best model from sequence

- The complexity of the regression models is optimized by choosing the optimal model in the sequential selection sequence.
- The process involves two steps.
  1. Fit statistics are calculated for the models that are generated in each step of the selection process on both training and validation data sets.



Best model from sequence

2. Then the simplest model, the model with the fewest inputs, with the optimal fit statistics on validation data is selected.



The role of regression models as predictive models

Common Applications of Multiple Regression Models

Multiple linear regression is a powerful tool for the following tasks:

- Prediction
  - To develop a model to predict future values of a response variable, Y, based on its relationships with other predictor or input variables, X.
- Analytical or Explanatory Analysis
  - To develop an understanding of the relationship between the response variable and predictor (input) variables.

Prediction

- The terms in the model, the values of their coefficients, and their statistical significance are of secondary importance.
- The focus is on producing a model that is the best at predicting future values of Y as a function of the Xs.
- The predicted value of Y is given by this formula:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1} X_1 + \cdots + \widehat{\beta_k} X_k$$

Analytical or Explanatory Analysis

- The focus is on understanding the relationship between the dependent variable and the independent variables.
- Consequently, the statistical significance of the coefficients, the magnitudes and signs of the coefficients are important.

$$\hat{Y} = \widehat{\boldsymbol{\beta_0}} + \widehat{\boldsymbol{\beta_1}} X_1 + \cdots + \widehat{\boldsymbol{\beta_k}} X_k$$

Lesson Summary

- The general linear model is one in which the target variable, Y, has a linear function of input variables, X. It agrees with the assumptions.
- The assumptions of general linear models are $\varepsilon \sim iid\ N(0, \sigma^2)$.
- Violation of the assumptions can affect various aspects of the model.
- All possible selection is the best input selection method, but the number of models to consider increases exponentially with the number of inputs.
- Sequential selection methods such as forward, stepwise and backward selection methods are the alternative to the exhaustive all possible selection.
- Sequential selection has its shortcomings.
- The optimal model is selected from the sequential selection sequence based on the best fit statistic on the validation data set.
- Multiple linear regression can be a prediction or analytical model. As a predictive model, the focus is on getting the best prediction value, Y.